

TABLEAU

Module 2 — Data Concepts

Topics Covered in This Session

1. Measures vs Dimensions
2. Discrete vs Continuous Data
3. Calculated Fields (with formulas + examples)
4. Parameters (with Dynamic Top N exercise)

Beginner-Friendly Notes with Real-Life Examples
Sample Dataset Included for Hands-On Practice

1. Measures vs Dimensions

Before you can build any chart in Tableau, you need to understand how it reads your data. Every field in your dataset is classified as either a Dimension or a Measure. This is the most fundamental concept in Tableau.

1.1 What is a Dimension?

A Dimension is a field that describes or categorises your data. Think of it as the 'label' or 'name' fields — they tell you WHO, WHAT, WHERE, or WHICH. It is denoted by 'Abc'

Simple Definition

Dimensions are the characteristics of your data. They are usually text, dates, or categories. You use them to SLICE or GROUP your data.

Real-life examples of Dimensions:

- **Customer Name** — e.g., 'Rahul Sharma', 'Priya Patel'
- **City / Region** — e.g., 'Mumbai', 'Delhi', 'West'
- **Product Category** — e.g., 'Electronics', 'Furniture', 'Clothing'
- **Order Date** — e.g., 15-Jan-2024
- **Gender** — e.g., 'Male', 'Female'

1.2 What is a Measure?

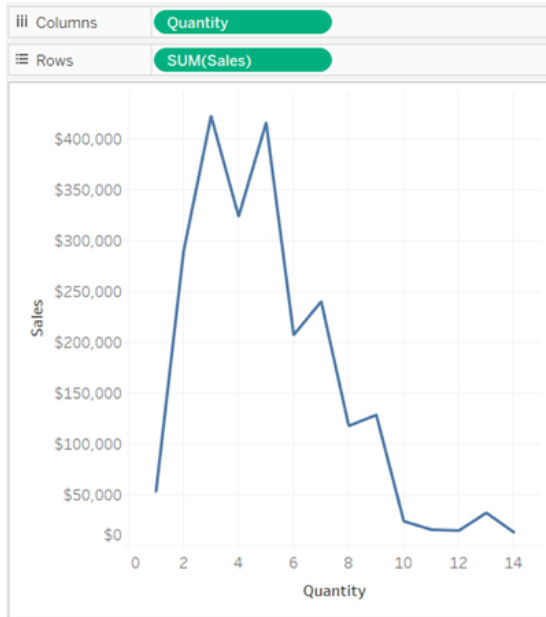
A Measure is a field that contains numbers you can calculate with — add, average, count, etc. Think of it as the 'amount' or 'score' fields. It is denoted by '#'

Simple Definition

Measures are the numbers in your data. They represent quantities you want to analyse — totals, averages, counts. You use them to CALCULATE.

Real-life examples of Measures:

- **Sales Amount** — e.g., ₹5,000 / ₹12,500
- **Profit** — e.g., ₹800 / -₹200
- **Quantity Sold** — e.g., 3 units / 10 units
- **Discount** — e.g., 10% / 0.15
- **Age of Customer** — e.g., 25, 42, 37



1.3 How Tableau Classifies Fields Automatically

When you connect Tableau to your data (Excel, CSV, database), it automatically sorts fields into two groups in the left panel:

- Text / String columns → placed under Dimensions (blue area at top)
- Date columns → placed under Dimensions
- Number columns → placed under Measures (green area at bottom)

⚠ Important Warning

Tableau is not always correct! Some numbers are NOT measures. For example: Zip Code (110001), Order ID (1001), Phone Number — these are numeric but should be Dimensions because you would never add or average them!

1.4 How to Convert Between Them

You can fix Tableau's classification easily:

1. Right-click the field in the Data panel on the left side
2. Click 'Convert to Dimension' or 'Convert to Measure'
3. Or simply drag the field from the Measures section up to the Dimensions section

1.5 Quick Reference Table

Concept	Real-Life Example
Dimension (Who/What/Where)	Customer Name → 'Rahul Sharma'
Dimension (Category)	Product Category → 'Electronics'
Dimension (Date)	Order Date → '15-Jan-2024'
Dimension (Geography)	City → 'Mumbai'
Measure (Amount)	Sales → ₹12,500
Measure (Count)	Quantity → 5 units
Measure (Ratio)	Profit Margin → 18.5%
Tricky! Convert to Dim	Zip Code → '400001' (do not sum!)

2. Discrete vs Continuous Data

This is where beginners often get confused. Even after you know Dimensions and Measures, there is another layer: whether a field is Discrete or Continuous. This controls how Tableau draws your chart axes.

2.1 What is Discrete Data?

Discrete = Separate / Countable Buckets

Discrete data has distinct, separate values with clear gaps between them. You can list them out individually. There is no 'in-between' value.

Real-life analogy: Think of the days of the week — Monday, Tuesday, Wednesday... There is nothing between Monday and Tuesday. You cannot have 'Monday and a half'. Each day is a separate, distinct category.

Examples of Discrete data:

- Months of the year: January, February, March... (12 distinct labels)
- Product categories: Electronics, Furniture, Clothing
- Number of children in a family: 0, 1, 2, 3 (whole numbers only)
- T-shirt sizes: S, M, L, XL

In Tableau:

- Discrete fields appear as BLUE pills on the shelves (Rows/Columns)
- They create HEADERS and LABELS on your chart (like column headings in a table)

2.2 What is Continuous Data?

Continuous = Flows Without Gaps / A Range

Continuous data can take any value within a range — including decimals and fractions. There are no gaps. It flows like a number line.

Real-life analogy: Think of temperature — it can be 22°C, 22.5°C, 22.7°C, 22.73°C... It flows continuously with infinite possible values between any two points.

Examples of Continuous data:

- Revenue: ₹1,000 to ₹1,00,000 (any decimal possible)
- Time / Duration: 2.5 hours, 3.7 minutes
- Weight: 55.3 kg, 72.8 kg
- A date range from Jan 1 to Dec 31 shown as a timeline

In Tableau:

- Continuous fields appear as GREEN pills on the shelves
- They create a NUMERIC AXIS (a ruler-like scale with min and max)

2.3 Blue vs Green Pills — Visual Guide

● BLUE Pill = Discrete	● GREEN Pill = Continuous
Creates column HEADERS / labels	Creates a numeric AXIS (like a ruler)
Shows data as separate buckets	Shows data as a flowing range
Example: Month names (Jan, Feb...)	Example: Sales axis (₹0 to ₹1,00,000)
Good for: Bar charts, comparisons	Good for: Line charts, trends over time

2.4 Practical Example — Sales by Month

Imagine you have monthly sales data. Here is the difference:

- **Discrete Month (Blue)** — Bar chart shows Jan | Feb | Mar as separate bars side-by-side. Each month is a labelled column.
- **Continuous Month (Green)** — Line chart flows smoothly from January to December along a timeline axis. You see the trend.

How to Switch

Right-click any pill that is already on the Rows or Columns shelf → Click 'Discrete' or 'Continuous'. Watch the pill change colour and the chart update instantly!

3. Calculated Fields

3.1 What is a Calculated Field?

Simple Definition

A Calculated Field is a new column you create inside Tableau using a formula. It does not change your original data — it is a virtual column that exists only in Tableau.

Real-life analogy: Imagine your Excel sheet has a 'Sales' column and a 'Cost' column, but no 'Profit' column. You can create a Calculated Field: $\text{Profit} = \text{Sales} - \text{Cost}$. Tableau will calculate this for every row automatically.

3.2 How to Create a Calculated Field

1. In Tableau, go to the top menu: Analysis → Create Calculated Field
2. Or right-click any empty space in the Data panel → Create Calculated Field
3. A dialog box opens. Give your field a name (e.g. 'Profit Margin')
4. Type your formula in the box below
5. Check the bottom-left: it should show a green tick and say 'The calculation is valid'
6. Click OK. Your new field appears in the Data panel with an '=' symbol

3.3 Our Practice Dataset

We will use this small sales dataset throughout all formula examples. Imagine this is data from a small electronics shop:

Order ID	Customer Name	Product	City	Sales (₹)	Cost (₹)	Quantity	Order Date	Returned
1001	Rahul Sharma	Laptop	Mumbai	55000	42000	1	05-Jan-2024	No
1002	Priya Patel	Mobile Phone	Delhi	18000	12000	2	10-Jan-2024	No
1003	Amit Verma	Headphones	Pune	3500	2000	3	12-Jan-2024	Yes
1004	Sneha Singh	Laptop	Mumbai	62000	48000	1	18-Jan-2024	No
1005	Kiran Kumar	Tablet	Chennai	25000	18000	1	22-Jan-2024	No
1006	Deepa Nair	Mobile Phone	Bangalore	16500	11000	1	25-Jan-2024	Yes
1007	Vikram Mehta	Headphones	Delhi	4200	2500	2	28-Jan-2024	No

Order ID	Customer Name	Product	City	Sales (₹)	Cost (₹)	Quantity	Order Date	Returned
1008	Sunita Reddy	Tablet	Hyderabad	28000	20000	1	30-Jan-2024	No

Profit Column is Missing!

Notice there is no 'Profit' column in the dataset. We will create it and many other useful fields using Calculated Fields below.

3.4 String Functions

String functions work with text fields. Let's use 'Customer Name' and 'City' from our dataset.

UPPER and LOWER

Converts text to ALL CAPS or all lowercase.

```
UPPER([Customer Name])
→ 'Rahul Sharma' becomes 'RAHUL SHARMA'
```

```
LOWER([City])
→ 'Mumbai' becomes 'mumbai'
```

When to use: Standardising data when some entries are mixed case — e.g., 'MUMBAI', 'mumbai', 'Mumbai' all become the same.

LEN — Length of Text

Counts how many characters are in a text field.

```
LEN([Customer Name])
→ 'Rahul Sharma' → 12
→ 'Priya Patel' → 11
```

CONTAINS

Returns TRUE if the text contains a specific word or character.

```
CONTAINS([Product], 'Phone')
→ 'Mobile Phone' → TRUE
→ 'Laptop' → FALSE
```

Use case: Create a filter to show only 'Phone' products across all product names.

LEFT, RIGHT, MID

Extract a portion of text from the start, end, or middle.

```
LEFT([Order ID], 2)      → '10'    (first 2 characters)
RIGHT([Customer Name], 5) → 'arma' (last 4 chars of 'Sharma')
MID([Customer Name], 1, 5) → 'Rahul' (5 chars starting at position 1)
```

3.5 Number Functions

Number functions work with numeric fields. We will use Sales, Cost, and Quantity.

Creating a Profit Field — Our First Calculated Field!

```
Field Name: Profit
Formula:    [Sales (₹)] - [Cost (₹)]

Results:
Rahul Sharma → 55000 - 42000 = 13000
Priya Patel  → 18000 - 12000 = 6000
Amit Verma   → 3500  - 2000  = 1500
```

ROUND

```
Field Name: Profit Margin %
Formula:    ROUND( ([Sales (₹)] - [Cost (₹)]) / [Sales (₹)] * 100 , 1 )

Results:
Rahul Sharma → ROUND(13000/55000*100, 1) = 23.6%
Priya Patel  → ROUND(6000/18000*100, 1)   = 33.3%
Amit Verma   → ROUND(1500/3500*100, 1)    = 42.9%
```

ABS — Absolute Value

Removes the minus sign from negative numbers.

```
ABS([Profit])
→ If profit is -2000 (a loss), ABS gives you 2000
→ Useful to show the SIZE of a loss without the negative sign
```

CEILING and FLOOR

```
CEILING(4.2) → 5      (always rounds UP to next whole number)
FLOOR(4.9)   → 4      (always rounds DOWN to previous whole number)
```

```
Example: CEILING([Quantity] * 1.1) → adds 10% buffer and rounds up
```

POWER

```
POWER([Quantity], 2) → squares the value
Quantity = 3          → POWER(3,2) = 9
```

3.6 Date Functions

Date functions work with the 'Order Date' field in our dataset.

TODAY and NOW

```
TODAY() → Returns today's date e.g. 24-May-2026
NOW()   → Returns current date + time e.g. 24-May-2026 14:30:00
```

DATEDIFF — Days Since Order

```
Field Name: Days Since Order
Formula:    DATEDIFF('day', [Order Date], TODAY())
```

Results (calculated from today, 24-May-2026):

```
Order 1001 (05-Jan-2024) → 505 days ago
Order 1005 (22-Jan-2024) → 488 days ago
```

Units you can use: 'day', 'month', 'year', 'hour', 'minute'

DATEPART — Extract Part of a Date

```
DATEPART('year', [Order Date]) → 2024
DATEPART('month', [Order Date]) → 1 (January = 1)
DATEPART('day', [Order Date]) → 5, 10, 12...
```

DATEADD — Add Time to a Date

```
Field Name: Expected Delivery
Formula:    DATEADD('day', 7, [Order Date])
```

Results:

```
Order 1001  05-Jan-2024 + 7 days → 12-Jan-2024
Order 1002  10-Jan-2024 + 7 days → 17-Jan-2024
```

3.7 Logical Functions

Logical functions let you create categories and labels based on conditions — like IF-THEN rules.

IF Statement — Categorise Sales

Field Name: Sales Category

Formula:

```
IF [Sales (₹)] >= 50000 THEN 'High Value'
ELSEIF [Sales (₹)] >= 15000 THEN 'Medium Value'
ELSE 'Low Value'
END
```

Results from our dataset:

```
Rahul Sharma    55000 → 'High Value'
Priya Patel     18000 → 'Medium Value'
Amit Verma      3500  → 'Low Value'
Sneha Singh     62000 → 'High Value'
Kiran Kumar     25000 → 'Medium Value'
```

IIF — Quick True/False

Field Name: Profit Status

Formula: IIF([Sales (₹)] - [Cost (₹)] > 0, 'Profitable', 'Loss')

All 8 orders in our dataset show: 'Profitable'

CASE — Rename Cities

Field Name: City Zone

Formula:

```
CASE [City]
  WHEN 'Mumbai' THEN 'West Zone'
  WHEN 'Pune'   THEN 'West Zone'
  WHEN 'Delhi'  THEN 'North Zone'
  WHEN 'Chennai' THEN 'South Zone'
  WHEN 'Bangalore' THEN 'South Zone'
  WHEN 'Hyderabad' THEN 'South Zone'
  ELSE 'Other Zone'
END
```

AND / OR / NOT — Multiple Conditions

Field Name: Premium Sale?

Formula: [Sales (₹)] > 20000 AND [Returned] = 'No'

Results:

Rahul Sharma	55000, No	→ TRUE (premium, not returned)
Priya Patel	18000, No	→ FALSE (sales not > 20000)
Amit Verma	3500, Yes	→ FALSE (both conditions fail)
Sneha Singh	62000, No	→ TRUE

3.8 Aggregate Functions

Aggregate functions calculate summary values across multiple rows. They are different from row-level functions — they give you ONE result for a group of rows.

Key Rule to Remember

You CANNOT mix aggregated and non-aggregated fields in one formula. For example: SUM([Sales]) - [Discount] will give an error. You must write: SUM([Sales]) - SUM([Discount]).

SUM([Sales (₹)])	→ Total Sales = ₹2,12,200 (all 8 orders combined)
AVG([Sales (₹)])	→ Average Sale = ₹26,525 per order
COUNT([Order ID])	→ 8 (counts all rows including duplicates)
COUNTD([City])	→ 6 (counts unique cities: Mumbai, Delhi, Pune, Chennai, Bangalore, Hyderabad)
MIN([Sales (₹)])	→ ₹3,500 (smallest order - Amit Verma)
MAX([Sales (₹)])	→ ₹62,000 (largest order - Sneha Singh)

4. Parameters

4.1 What is a Parameter?

Simple Definition

A Parameter is like a dial or slider you put on your dashboard that the viewer can adjust. When they move it, the chart updates instantly. It lets users interact with the data without you having to rebuild anything.

Real-life analogy: Think of a volume knob on a speaker. You turn it up or down, and the sound changes. A Parameter is exactly that — but for your Tableau charts.

The difference between a Filter and a Parameter:

Filter	Parameter
Tied to a specific field	Not tied to any field — holds a single value
Shows/hides rows of data	Changes a number or value used in a formula
Example: Filter City = Mumbai	Example: Set Top N = 5
Fixed options from your data	You define the range and default value

4.2 How to Create a Parameter

1. In the Data panel, click the small dropdown arrow at the top right → Create Parameter
2. A dialog opens. Fill in:
 - a. Name: Give it a clear name, e.g. 'Top N Customers'
 - b. Data Type: Choose Integer (for a whole number like 5, 10, 20)
 - c. Current Value: This is the default — set it to 5
 - d. Allowable Values: Choose 'Range'
 - e. Minimum: 1 | Maximum: 8 | Step Size: 1
3. Click OK — the Parameter appears at the bottom of the Data panel
4. Right-click it → 'Show Parameter Control' — a slider appears on the right side of the view

4.3 Using a Parameter in a Calculated Field

A Parameter does nothing on its own. You must connect it to a Calculated Field that uses the parameter's value.

Reference the parameter in a formula using square brackets:

`RANK( SUM([Sales (₹)]) ) <= [Top N Customers]`

This formula returns TRUE for customers ranked 1 through N and FALSE for everyone else.

4.4 Practical Example — Dynamic Top N Customers

Goal

Build a bar chart that shows Top 5 customers by sales. The user can change the slider to see Top 3, Top 5, or Top 8 — and the chart updates automatically.

Our Dataset — 8 Customers Ranked by Sales:

Rank	Customer Name	City	Sales (₹)	In Top 5?
1	Sneha Singh	Mumbai	₹62,000	✓ YES
2	Rahul Sharma	Mumbai	₹55,000	✓ YES
3	Kiran Kumar	Chennai	₹25,000	✓ YES
4	Sunita Reddy	Hyderabad	₹28,000	✓ YES
5	Priya Patel	Delhi	₹18,000	✓ YES
6	Deepa Nair	Bangalore	₹16,500	✗ No
7	Vikram Mehta	Delhi	₹4,200	✗ No
8	Amit Verma	Pune	₹3,500	✗ No

Step-by-Step Instructions:

STEP 1 Create the Parameter	Name: Top N Customers Type: Integer Default: 5 Range: 1 to 8 Step: 1
STEP 2 Create a Calculated Field	Name: 'Top N Filter' Formula: <code>RANK(SUM([Sales (₹)])) <= [Top N Customers]</code> This returns TRUE for the top N customers by sales
STEP 3 Build the chart	Drag Customer Name to the Rows shelf Drag Sales (₹) to the Columns shelf Sort the bars: click the sort icon → Descending by Sales

STEP 4 Apply the filter	Drag 'Top N Filter' to the Filters shelf In the filter dialog, check only 'True' → Click OK
STEP 5 Show the control	Right-click 'Top N Customers' in the Data panel Click 'Show Parameter Control' A slider appears on the right side of your view
STEP 6 Test it!	Move the slider from 3 to 8 Watch the bar chart grow and shrink as you add/remove customers Your dynamic Top N chart is complete!

What You Have Built

A professional interactive chart where the business user can drag a slider and instantly see the Top 3, Top 5, or Top 8 customers. This is the power of Parameters — no code, no rebuilding, just drag and interact!

Module 2 — Quick Reference Cheat Sheet

Measures vs Dimensions

Dimension	Measure
Describes / categorises data	Quantifies / calculates data
Text, dates, geography	Numbers you aggregate
BLUE area in Data panel	GREEN area in Data panel
Examples: Name, City, Category, Date	Examples: Sales, Profit, Quantity

Discrete vs Continuous

Discrete (Blue pill)	Continuous (Green pill)
Separate, countable values	Flows along a range
Creates headers/labels	Creates numeric axis
Good for comparisons	Good for trends
Example: Month = Jan, Feb, Mar	Example: Revenue = ₹0 to ₹1,00,000

All Calculated Field Functions — Summary

Category	Function	Example	Result
String	UPPER	UPPER('mumbai')	'MUMBAI'
String	LEN	LEN('Rahul')	5
String	LEFT	LEFT('Laptop', 3)	'Lap'
String	CONTAINS	CONTAINS('Mobile Phone','Phone')	TRUE
Number	ROUND	ROUND(23.567, 1)	23.6
Number	ABS	ABS(-1500)	1500
Number	CEILING	CEILING(4.2)	5
Date	DATEDIFF	DATEDIFF('day', OrderDate, TODAY())	Days elapsed
Date	DATEPART	DATEPART('month', OrderDate)	1 for January
Date	DATEADD	DATEADD('day', 7, OrderDate)	OrderDate + 7 days
Logical	IF/END	IF Sales>50000 THEN 'High' END	'High'

Category	Function	Example	Result
Logical	IIF	IIF(Profit>0, 'Win', 'Loss')	'Win'
Logical	CASE	CASE City WHEN 'Mumbai' THEN...	Grouped zones
Aggregate	SUM	SUM([Sales])	Total sales
Aggregate	COUNTD	COUNTD([City])	Unique cities
Aggregate	AVG	AVG([Sales])	Average sales

Parameter — Steps at a Glance

1	Data panel → dropdown → Create Parameter
2	Name it, set Type (Integer), set Range (min/max/step), set default
3	Create Calculated Field that references [Parameter Name]
4	Use the calculated field as a Filter → select TRUE
5	Right-click parameter → Show Parameter Control